



Chapter 7: DC-DC Switch-Mode Converters:

7-1 Introduction

7-2 Control of dc-dc Converters

7-3 Step-Down (Buck) Converter

7-4 Step-Up (Boost) Converter

7-5 Buck-Boost Converter

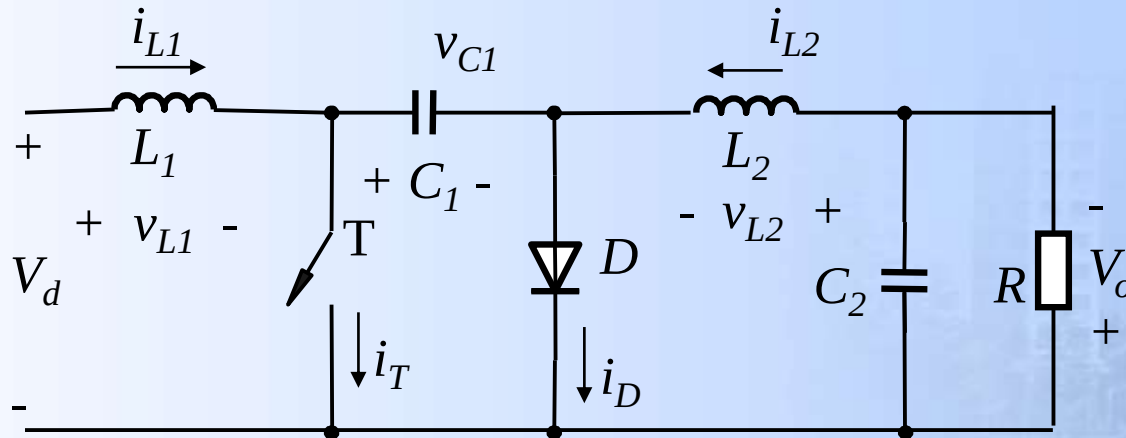
7-6 Cuk dc-dc Converter

7-7 Full Bridge dc-dc Converter

7-8 dc-dc Converter Comparison



7-6 Cuk dc-dc Converter



- Similar to a buck-Boost converter, the cuk converter provides a **negative-polarity output voltage** with respect to the common terminal of the input voltage, and the output voltage can be either higher or lower than the input voltage.
- unlike the buck-boost converter, both the input current and the output current are not pulsation.
- In steady-state: $V_{c1} = V_d + V_o$



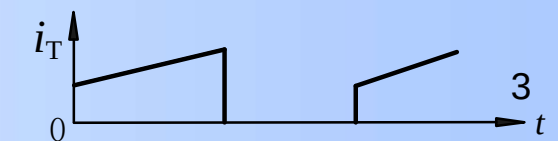
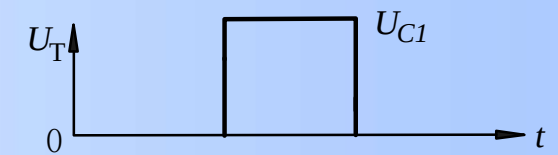
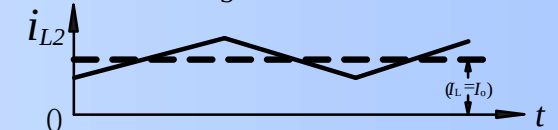
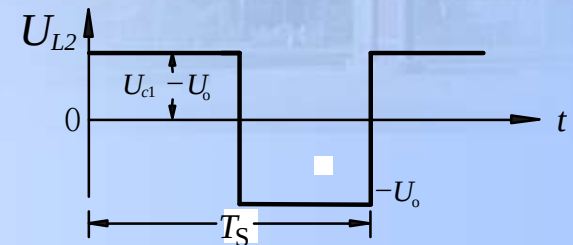
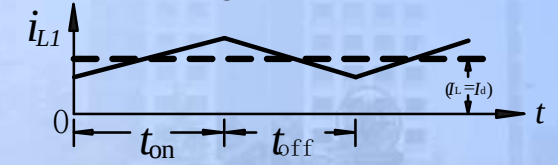
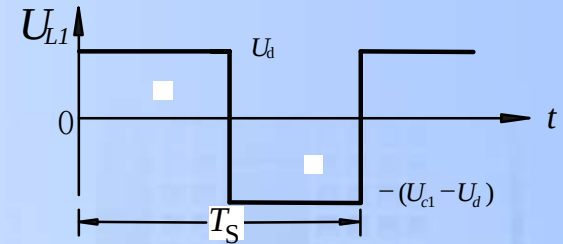
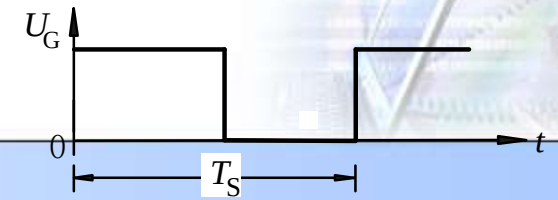
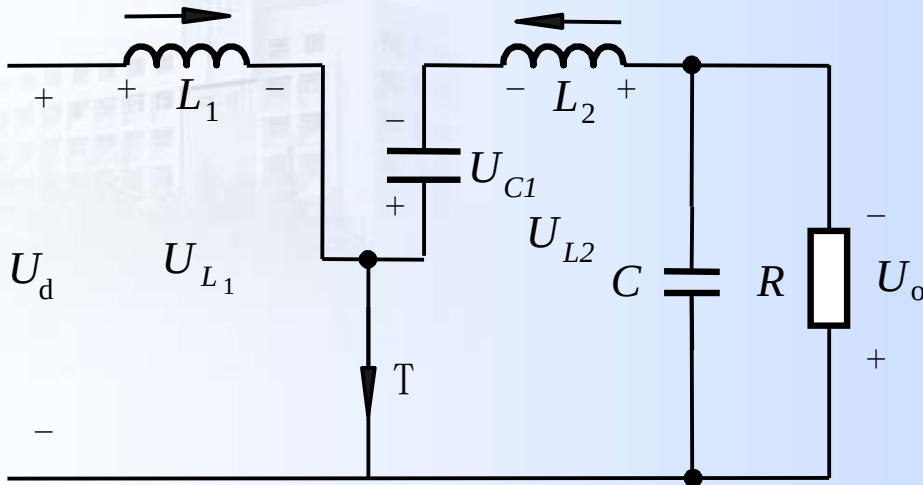
7-6-1 continuous-conduction mode

☆ 1 Operation principle

(1) When the switch is on:

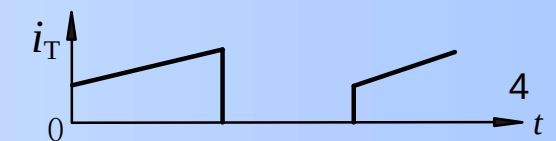
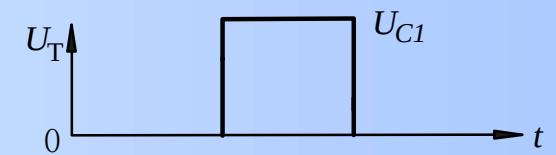
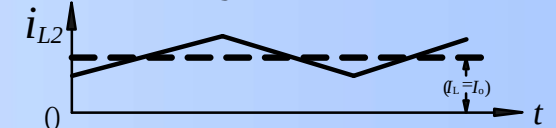
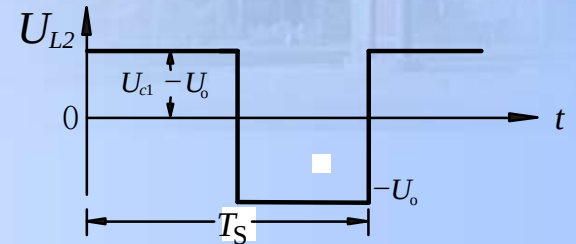
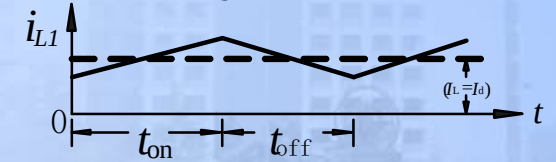
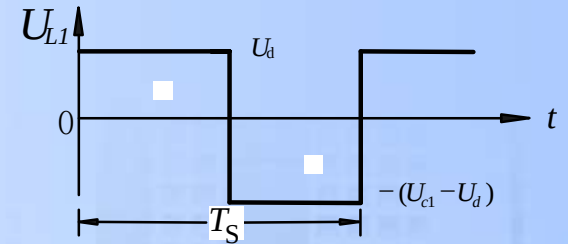
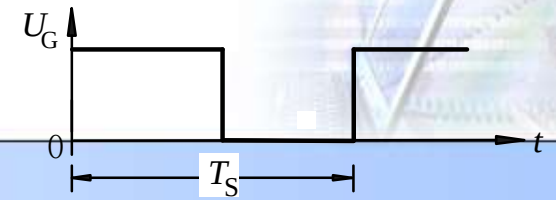
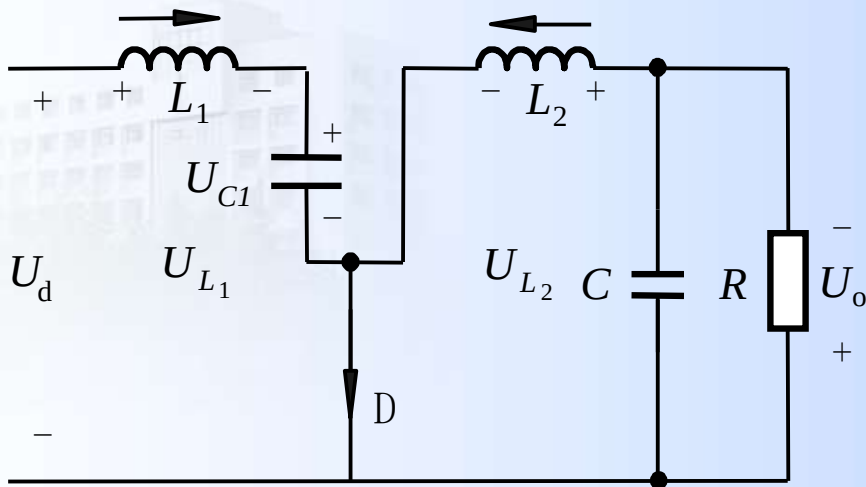
The input feeds energy to L_1 , $i_{L1} \uparrow$.

V_{C1} reverse biases the diode. i_{L1} and i_{L2} flow through the switch. C_1 discharge through the switch, transferring energy to the output and L_2 . $i_{L2} \uparrow$.





(2) When the switch is off:
 i_{L1} and i_{L2} flow through the diode. C_1 is charged by energy from both the input and L_1 , $i_{L1} \downarrow$. L_2 feeds the output, $i_{L2} \downarrow$.





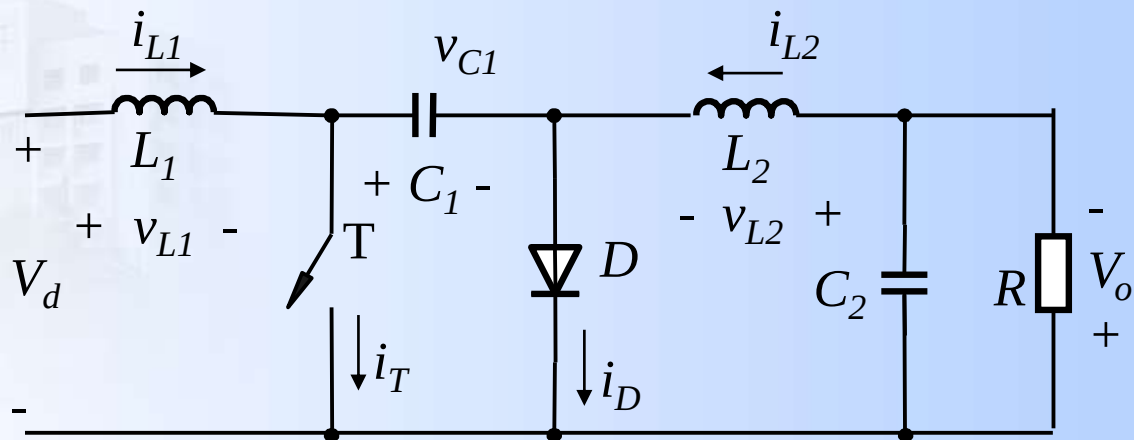
2 Steady-state analysis:

In the dc steady state:

$$V_{L1} = 0 \quad V_{L2} = 0$$

$$I_{C1} = 0 \quad I_{C2} = 0$$

$$I_d = I_{L1} \quad I_o = I_{L2}$$





$$L_1: \quad V_d D T_s + (V_d - V_{C1})(1-D)T_s = 0$$

$$\therefore V_{C1} = \frac{1}{1-D} V_d \quad (7-59)$$

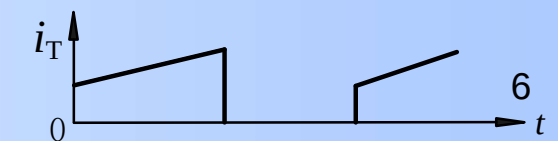
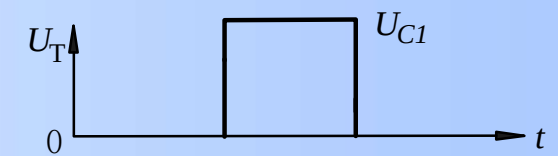
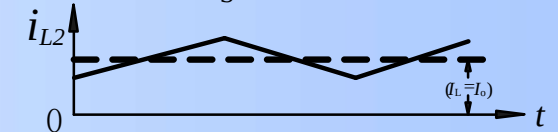
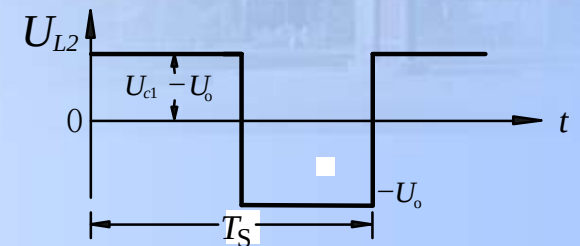
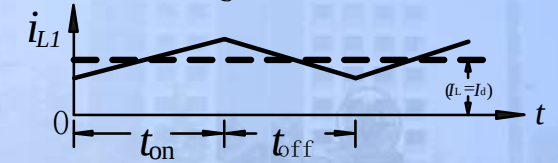
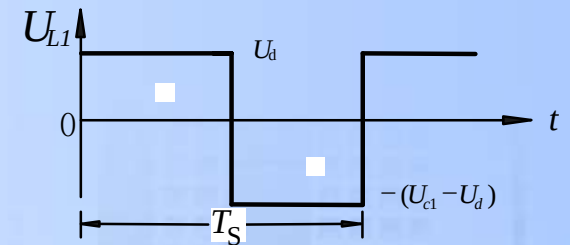
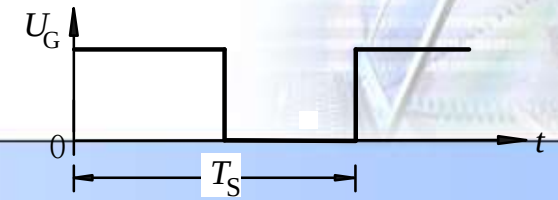
$$L_2: \quad (V_{C1} - V_o) D T_s + (-V_o)(1-D)T_s = 0$$

$$\therefore V_{C1} = \frac{1}{D} V_o \quad (7-60)$$

$$\therefore \frac{V_o}{V_d} = \frac{D}{1-D} \quad (7-61)$$

$$\frac{I_o}{I_d} = \frac{1-D}{D} \quad (7-62)$$

➤ V_o can be controlled by varying the duty ratio D .





Example 7-3

In a Cuk converter operating at 50kHz, $L_1=L_2=1\text{mH}$ and $C_1=5\mu\text{F}$. The output capacitor is sufficiently large to yield an essentially constant output voltage. Here $V_d=10\text{V}$ and the output V_o is regulated to be constant at 5V. It is supplying 5W to a load. Assume ideal components. Calculate the percentage errors in assuming a constant voltage across C_1 or in assuming constant current i_{L1} and i_{L2} .

Solution

(a) If the voltage across C_1 is assumed to be constant, from Eq. 7-58: $v_{C1}=V_{C1}=V_d+V_o=10+5=15\text{V}$.

Initially, we will assume the current conduction to be continuous. Therefore, from Eq. 7-61: $D/(1-D)=5/10$, $D=0.33$.



During the off interval:

$$\Delta i_{L1} = \frac{V_{C1} - V_d}{L_1} (1 - D) T_s = \frac{15 - 10}{10^{-3}} (1 - 0.333) \times 20 \times 10^{-6} = 0.067(A)$$

$$\Delta i_{L2} = \frac{V_o}{L_2} (1 - D) T_s = \frac{5}{10^{-3}} (1 - 0.333) \times 20 \times 10^{-6} = 0.067(A)$$

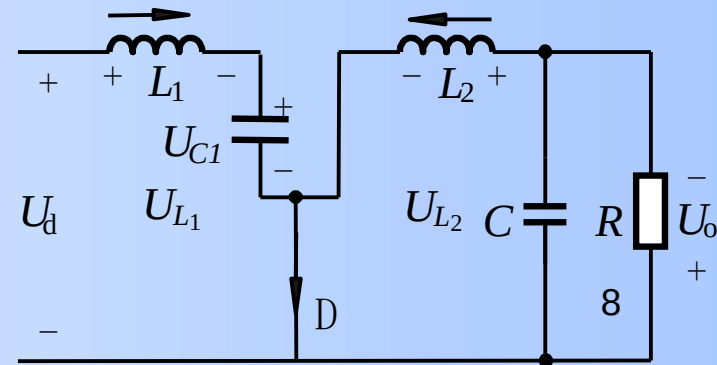
At an output load of 5W: $I_o = P_o / V_o = 5/5 = 1A$ and $I_d = 0.5A$.

Since $\Delta i_{L1} < I_d$ (I_{L1}) and $\Delta i_{L2} < I_o$ (I_{L2}), the mod of operation is **continuous**, as assumed earlier.

Therefore, the percentage errors in assuming constant I_{L1} and I_{L2} would be:

$$\frac{\Delta i_{L1}}{I_{L1}} = \frac{0.067 \times 100}{0.5} = 13.4\%$$

$$\frac{\Delta i_{L2}}{I_{L2}} = \frac{0.067 \times 100}{1} = 6.7\%$$



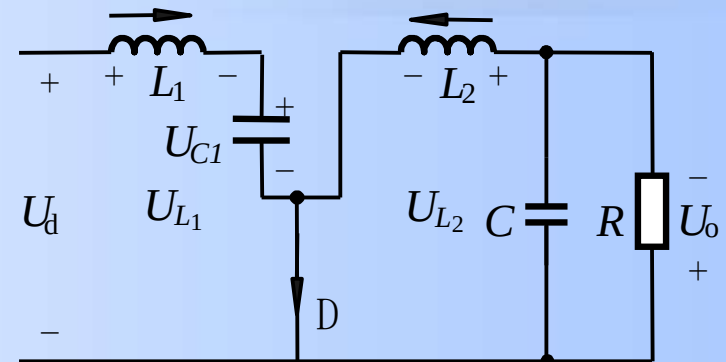


Solution

(b) If i_{L1} and i_{L2} are assumed to be constant, during the off interval:

$$\Delta V_{C1} = \frac{1}{C} \int_0^{(1-D)T_s} i_{L1} dt \approx \frac{I_{L1}}{C} (1-D)T_s = \frac{0.5 \times (1-0.333) \times 20 \times 10^{-6}}{5 \times 10^{-6}} = 1.33(\text{V})$$

$$\frac{\Delta V_{C1}}{V_{C1}} = \frac{1.33 \times 100}{15} = 8.87\%$$





Chapter 27: Snubber Circuits

27-1 Induction

27-2 Diodes Snubber

27-3 Snubber Circuits for Thyristors

27-4 Need for Snubbers with Transistors

27-5 Turn-off Snubber

27-6 Overvoltage snubber

27-7 Turn-on Snubber



知识点:

- 1、掌握二极管与晶闸管、全控开关器件的缓冲电路的特性以及实现形式;
- 2、掌握开通缓冲电路、过压缓冲电路以及关断缓冲电路的特性以及实现形式。

27-1 Induction

1 The issues which the switch faced in applications (电力电子器件应用时面临的问题)

(1) switching loss

$$P_s = \frac{V_d I_o f_s (t_{c(on)} + t_{c(off)})}{2}$$

→ Switching loss $\propto f_s$, we want $f_s \uparrow$ So L, C size $\downarrow$, but losses limit us.

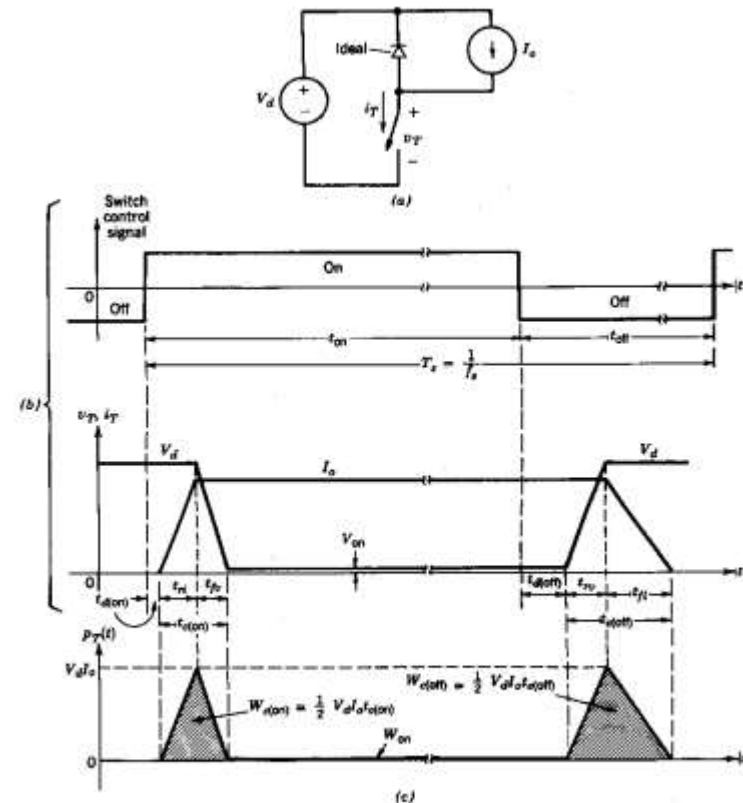
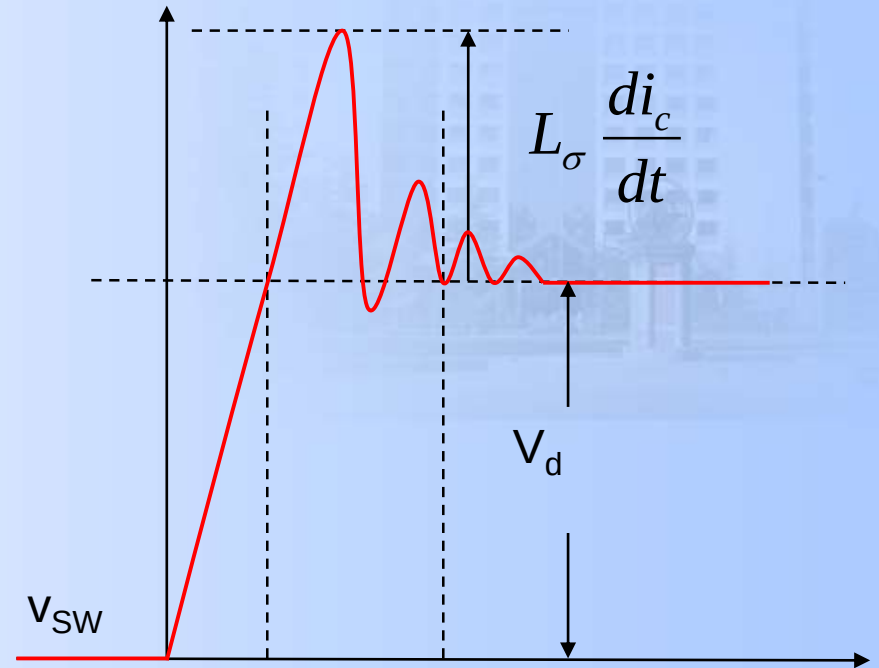
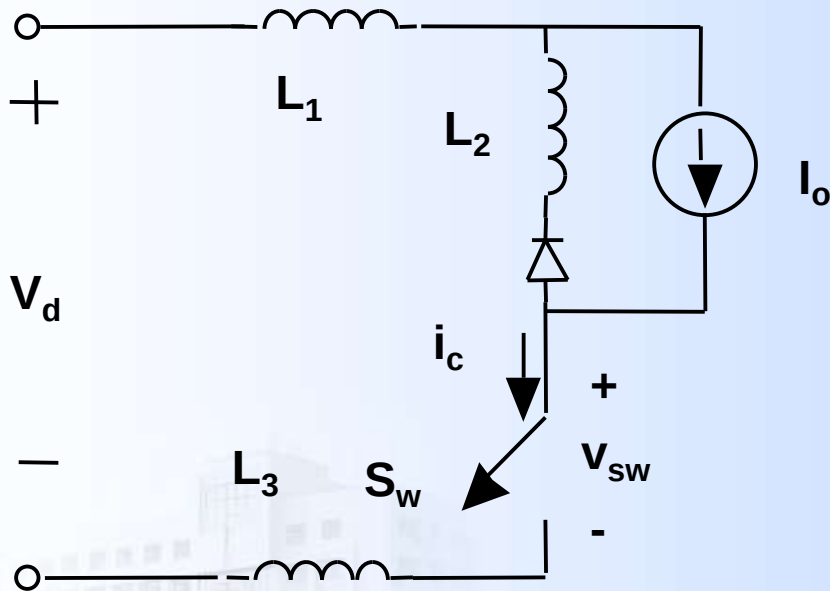


Figure 2-6 Generic-switch switching characteristics (linearized): (a) simplified clamped-inductive-switching circuit, (b) switch waveforms, (c) instantaneous switch power loss.



(2) Overshoots (电压过冲)



$L_1, L_2, L_3 =$ stray inductance (杂散电感)

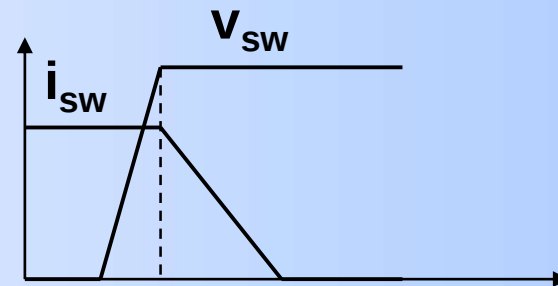
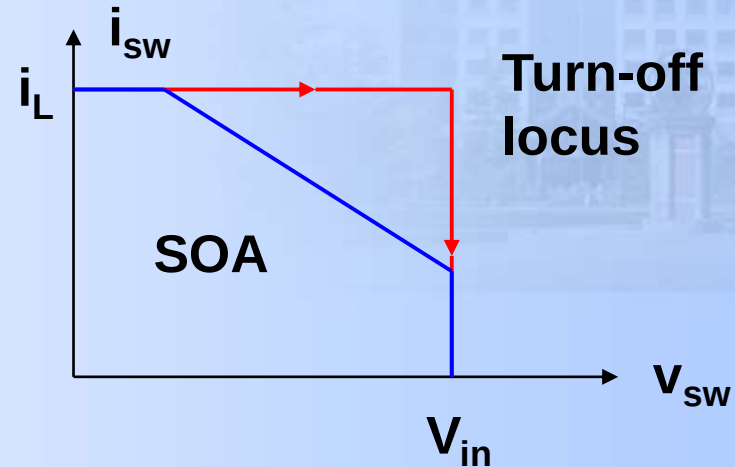
$$L_\sigma = L_1 + L_2 + L_3$$



(3) Safe operating area (SOA)

Some devices have limitations on simultaneous voltage + current applied instantaneously (IGBT, MOSFET, etc.)

Device can operate at I_L and can operate at V_{in} , but not at same time





(4) di/dt or dv/dt limits

Also, some devices have di/dt or dv/dt limits (esp. thyristor type devices).

(5) EMI

High di/dt , dv/dt lead to EMI, affects control circuits.
($dv/dt \rightarrow$ capacitive coupling, $di/dt \rightarrow$ inductive coupling)



2 The ways which the switch being overstressed in applications (电力电子器件承受的过度冲击的方式/类型)

Five possibilities for overstresses:

at turn-off

① Overvoltage and ② high dv/dt

at turn-on

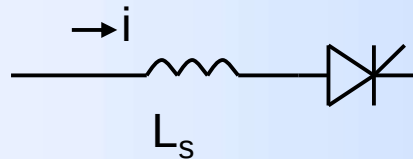
③ Overcurrent and ④ high di/dt ,

other

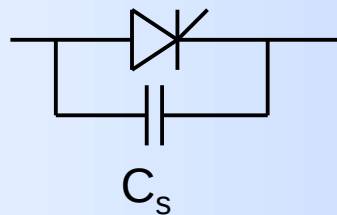
⑤ Excessive power dissipation.



➤ Snubber circuit can be added to the basic converter to reduce the stresses to safe levels.



The di/dt is limited by adding a series inductor L_s



The dv/dt is limited by connecting a capacitor C_s



3 Snubber's Function ☆ : Protect semiconductor devices by:

- Shaping the switching trajectory of the device as it turns on/off and lows the **device dissipation**.

(改变器件开通/关断轨迹来降低器件的开关损耗)

- Limiting device voltages during turn-off transients
- Limiting the rate-of-rise (dv/dt) of voltages across the semiconductor device at device **turn-off**.

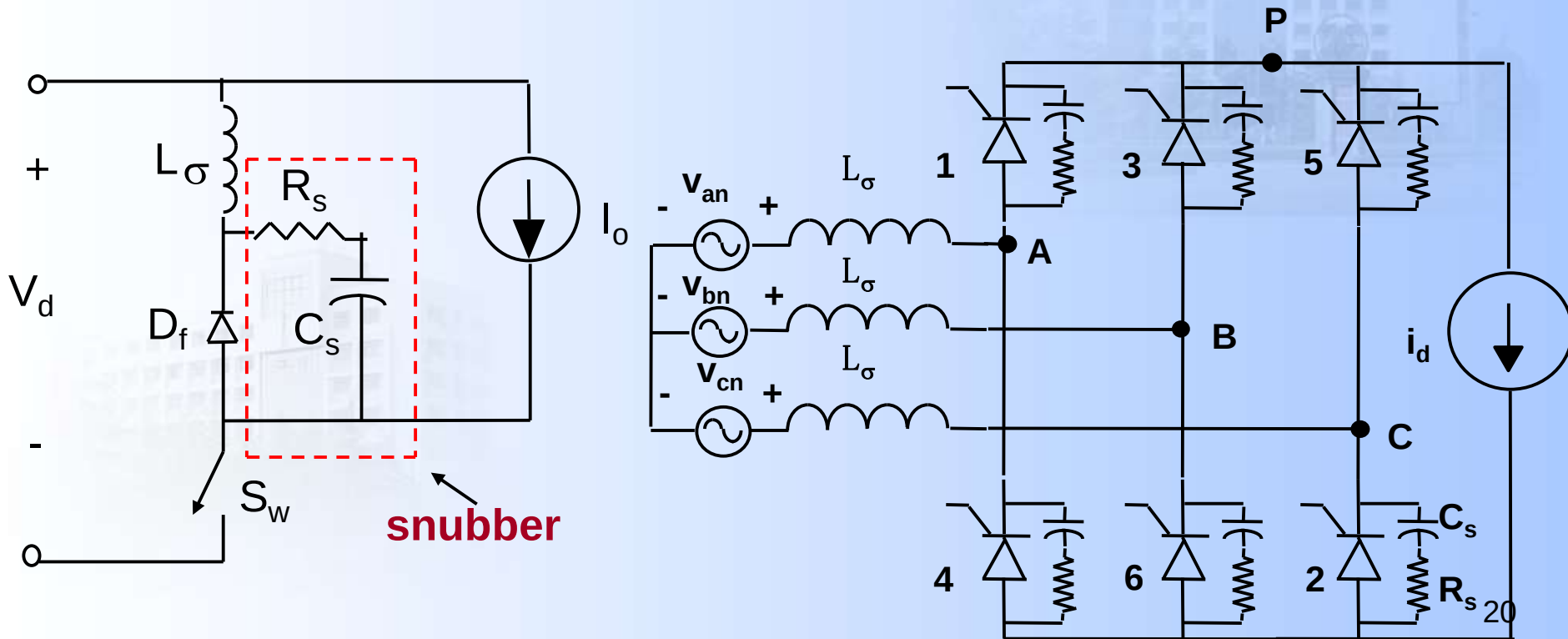
(限制器件关断期间承受的电压和承受的 dv/dt)



3 Snubber's Function ☆ : Protect semiconductor devices by:

- Limiting device currents during turn-on transients
- Limiting the rate-of-rise (di/dt) of currents through the semiconductor device at device turn-on

(限制器件开通期间承受的电流和承受的 di/dt)



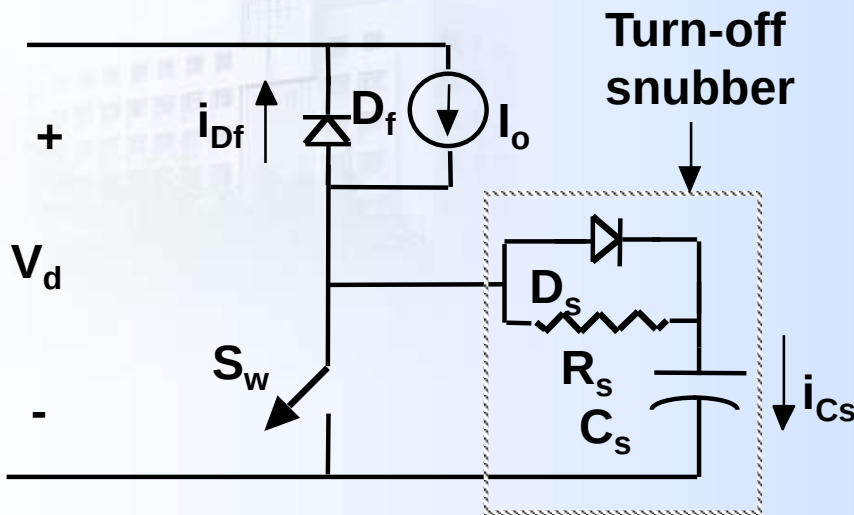


(2) Polarized R-C snubbers (有极性RC缓冲电路)

- Used as turn-off snubbers to shape the turn-off switching trajectory of controlled switches.
- Used as overvoltage snubbers to clamp voltages applied to controlled switches to safe values.
- Limit dv/dt during device turn-off

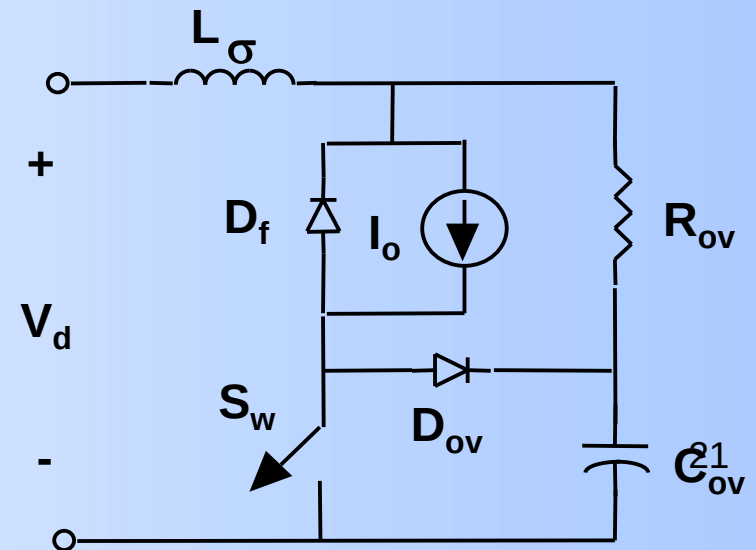
➤ turn-off snubbers

(关断Snubber)



➤ Overvoltage Snubber

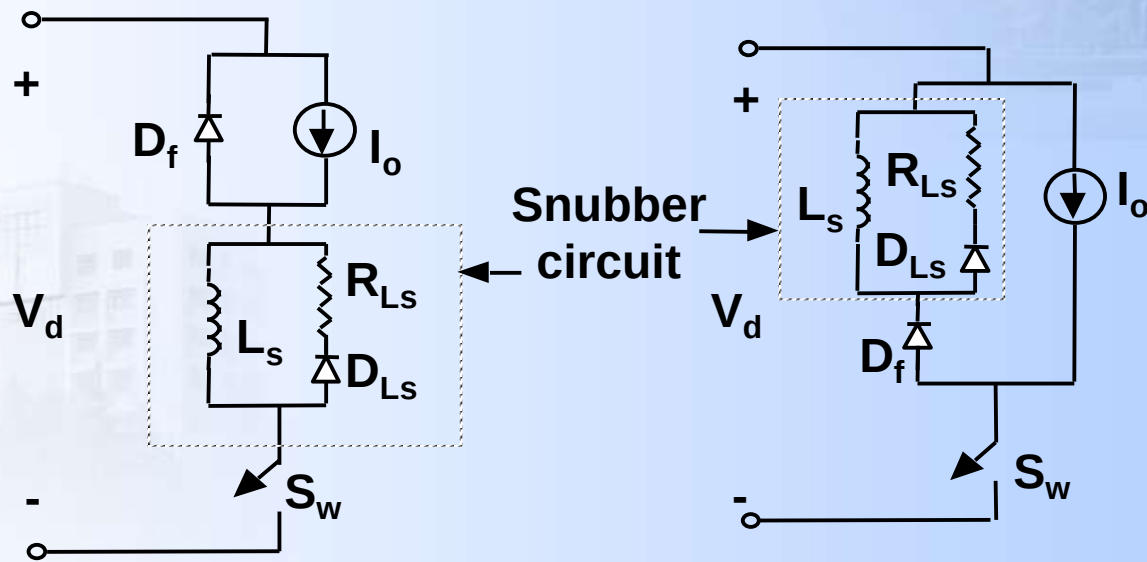
(过电压Snubber)





(3) Polarized L-R snubbers (有极性LR缓冲电路)

- Used as turn-on snubbers to shape the turn-on switching trajectory of controlled switches.
- Limit di/dt during device turn-on





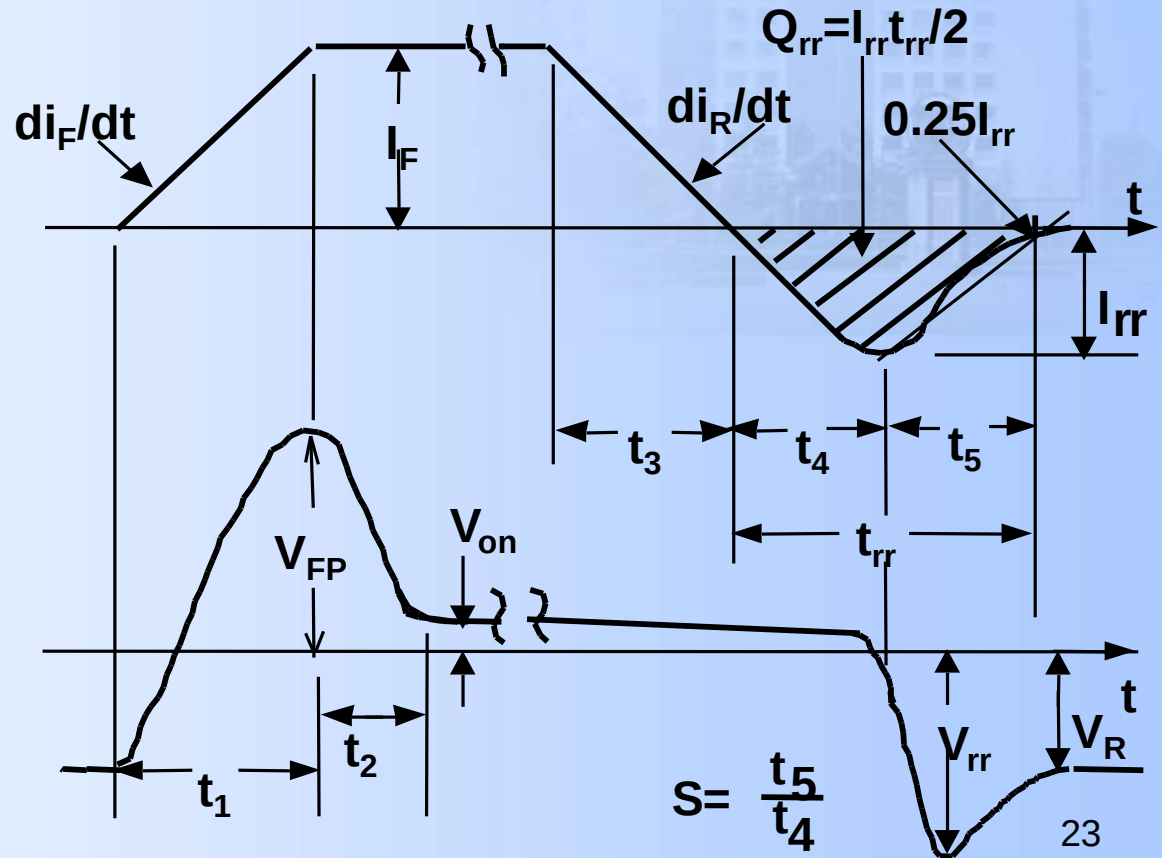
1 Switching Waveforms

A power diode requires a finite time to switch from blocking state to the on state (forward bias) and vice versa.

Forward-recovery voltage (正向恢复峰值电压) V_{FP}

The overshoot of the voltage during turn-on is not observed.

Forward-recovery time (正向恢复时间) ($t_{fr} = t_1 + t_2$)



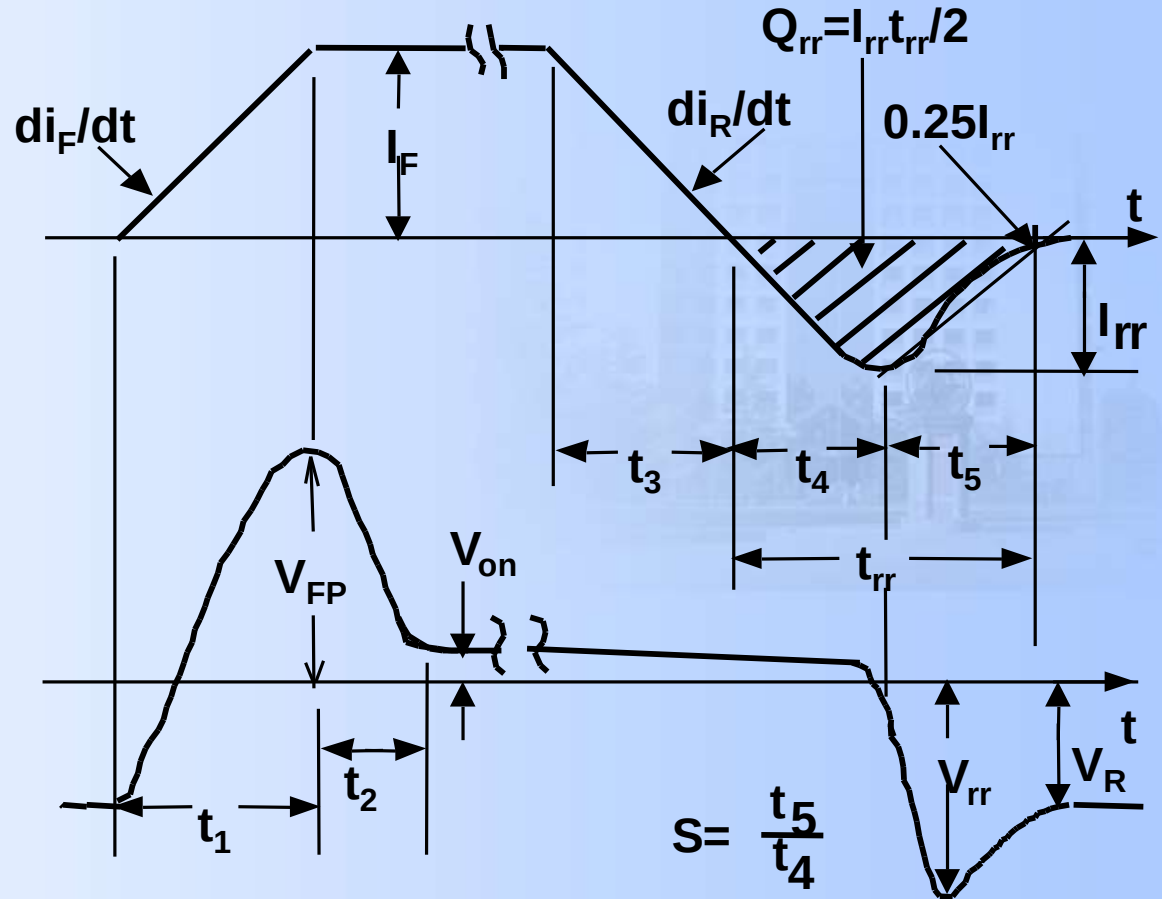


27-2 Diode Snubber Circuit

Reverse-recovery voltage(反向恢复峰值电压) V_{rr}

Reverse-recovery current(反向恢复峰值电流) I_{rr}

Reverse-recovery time(反向恢复时间) t_{rr}

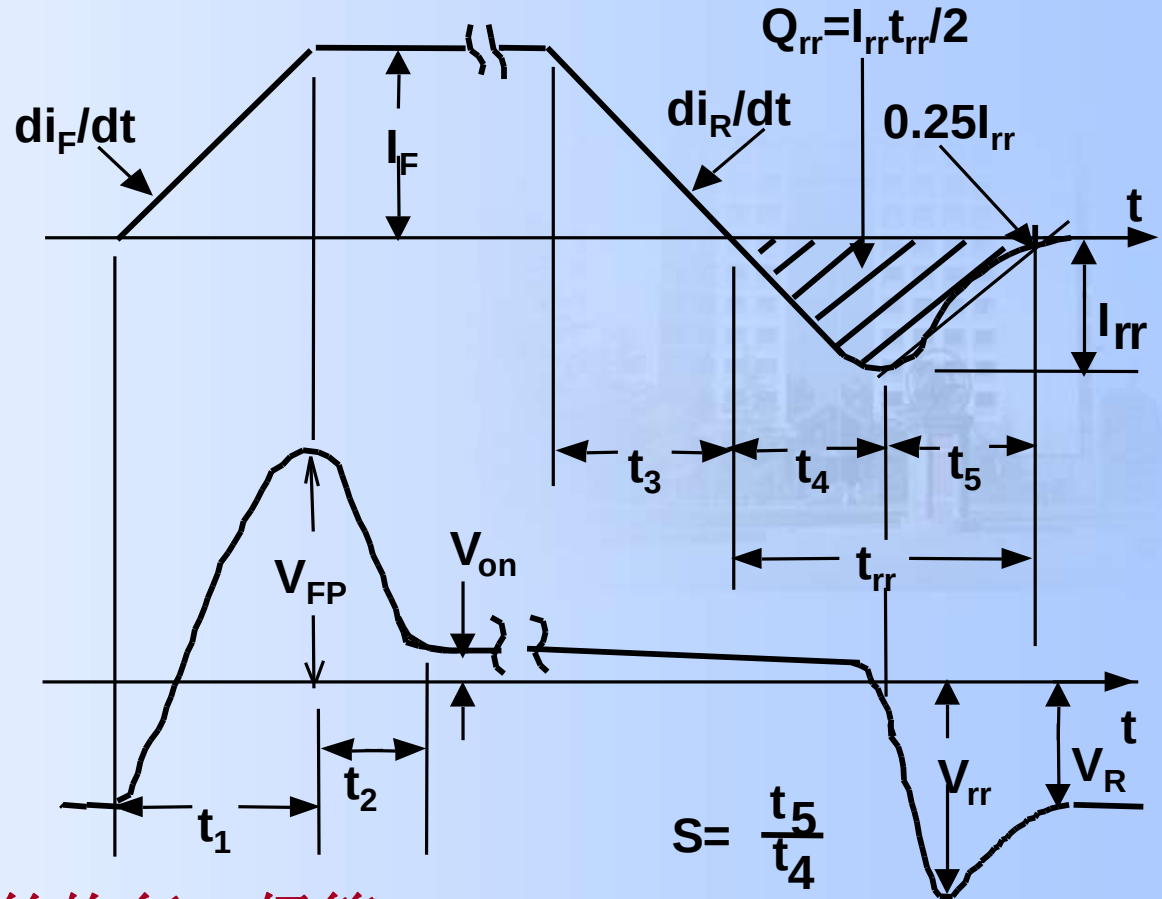




27-2 Diode Snubber Circuit

Reverse-recovery charge (反向恢复电荷) $Q_{rr} = I_{rr} t_{rr} / 2$

“snappiness” factor (恢复系数)
 $S = t_5 / t_4$



Soft-recovery diode (软恢复二极管): $t_5 > t_4$

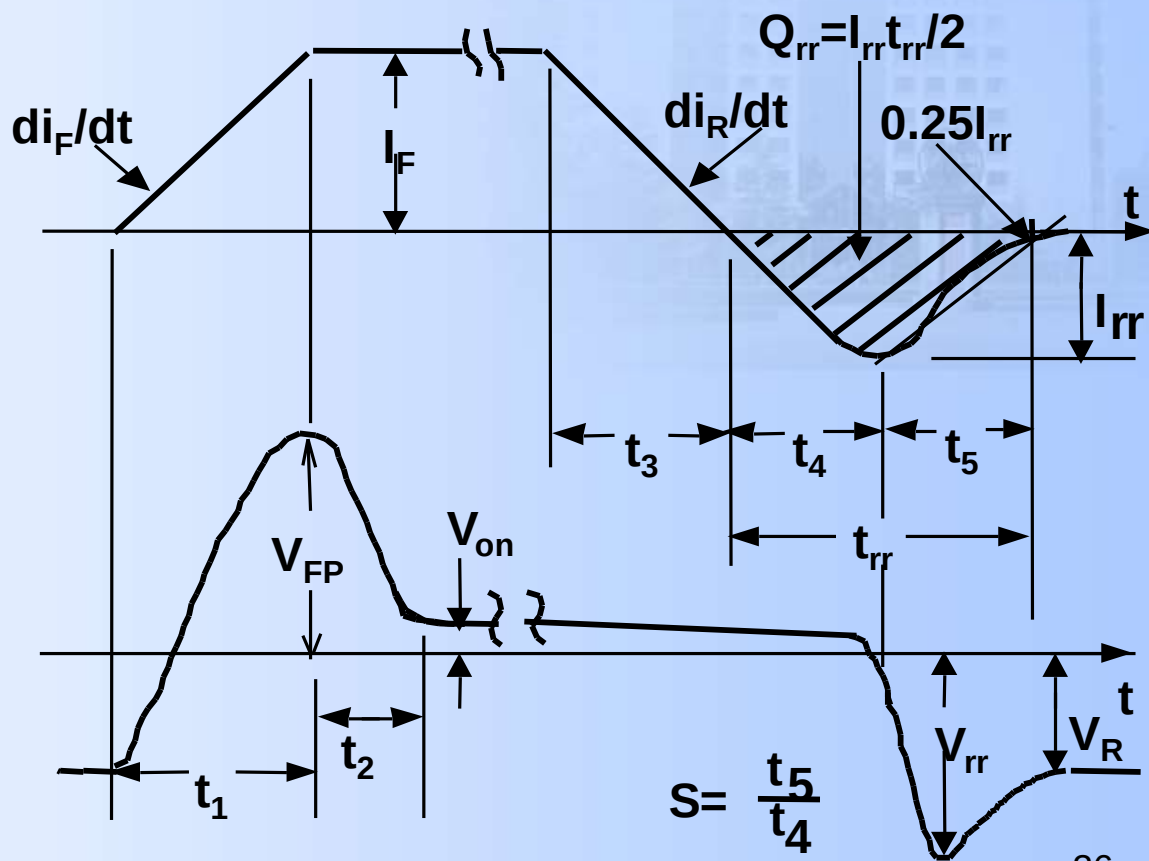
Abrupt-recovery diode (硬恢复二极管): $t_5 \ll t_4$



27-2 Diode Snubber Circuit

- di_F/dt and di_R/dt are determined by external circuit.

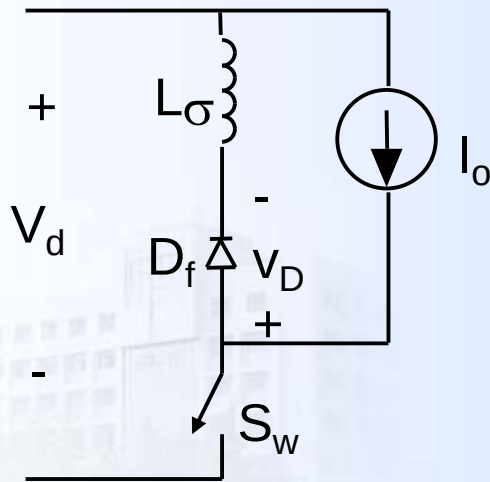
$$I_{rr} = \frac{di_R}{dt} t_4 = \frac{di_R}{dt} \frac{t_{rr}}{S+1}$$



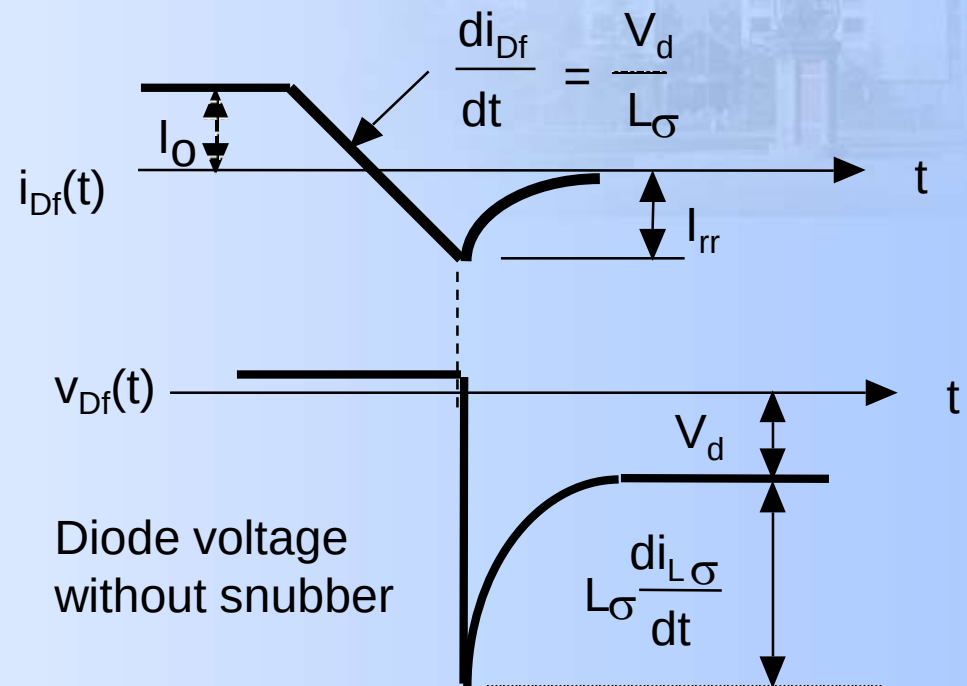


2 The cause of overvoltages

The overvoltages occur due to the **stray**(杂散电感) or **leakage inductance** L_σ in series with the diode and the snap-off of the diode reverse recovery current.



$L_\sigma =$ stray inductance





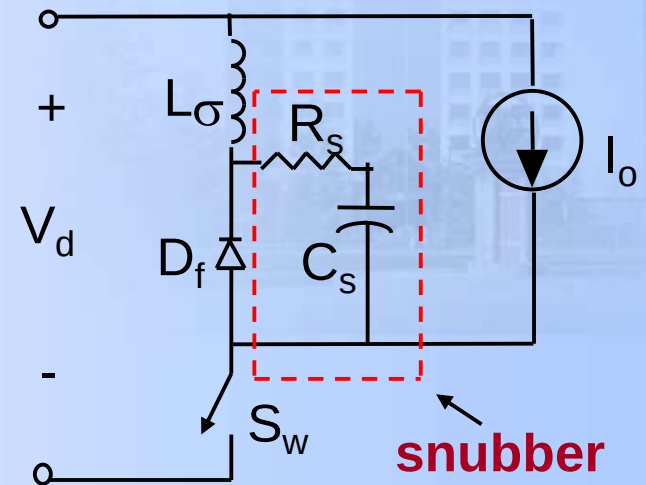
3 overvoltage protection

Introducing a R_s - C_s snubber circuit

The roles of elements

C_s 的作用是限制 D_f 上的 dv/dt 和关断过电压；

R_s 的作用是限制 C_s 的放电电流。

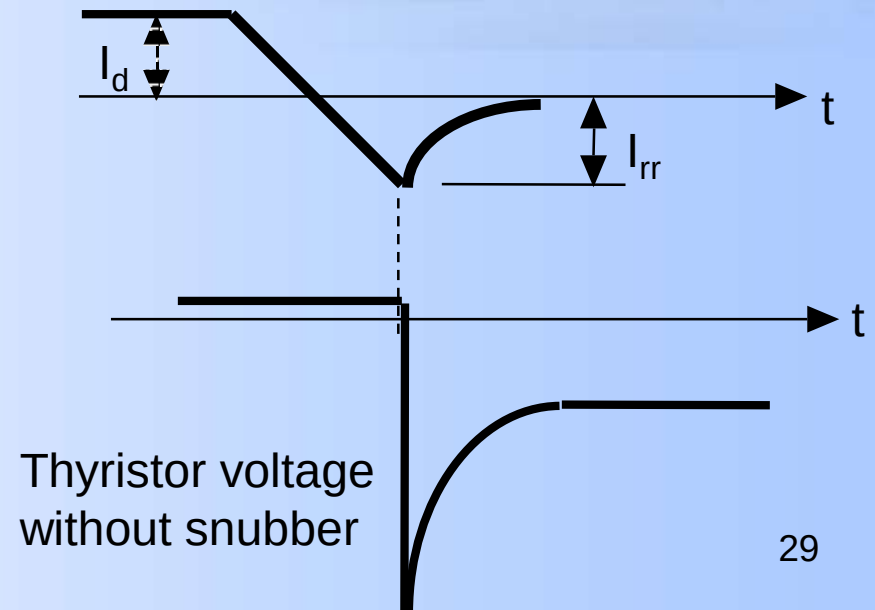
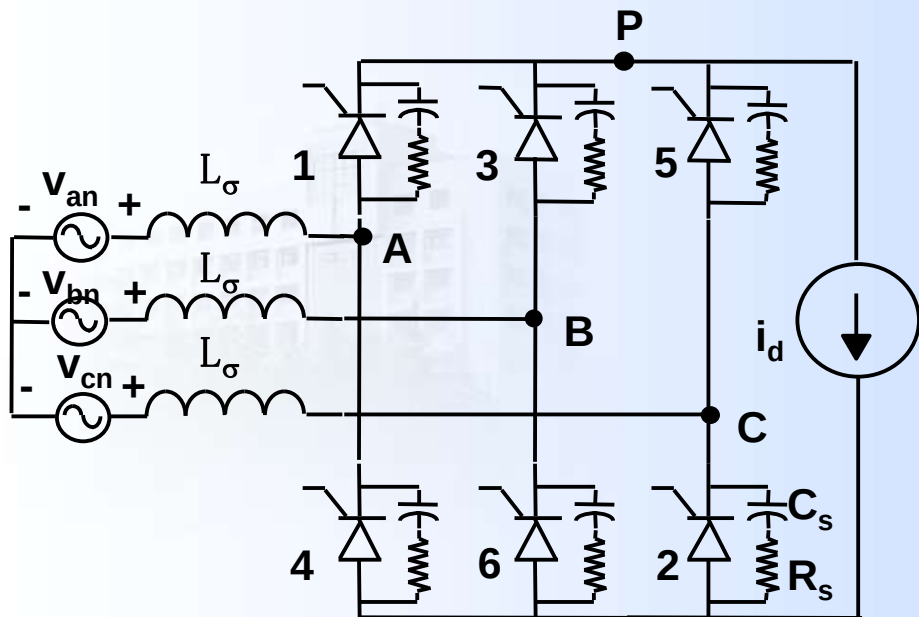




27-3 Snubber Circuit for Thyristors

The cause of overvoltages

The overvoltages occur due to the **series inductance L_σ** (line reactance + transformer leakage inductance) and the **snap-off of the thyristor reverse recovery current**.





27-4 Need for Snubbers with Controlled Switches

Snubber circuits are used to protect the transistors by improving their switching trajectory. Three basic types of snubbers:

- 1 Turn-off snubbers
- 2 Turn-on snubbers
- 3 Overvoltage snubbers

Step-down converter example

$L_1, L_2, L_3 =$ stray inductance

$$L_{\sigma} = L_1 + L_2 + L_3$$

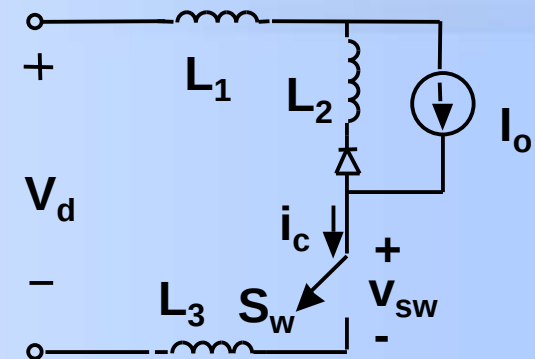


Fig. 27-11(a)



Switch current and voltage waveforms

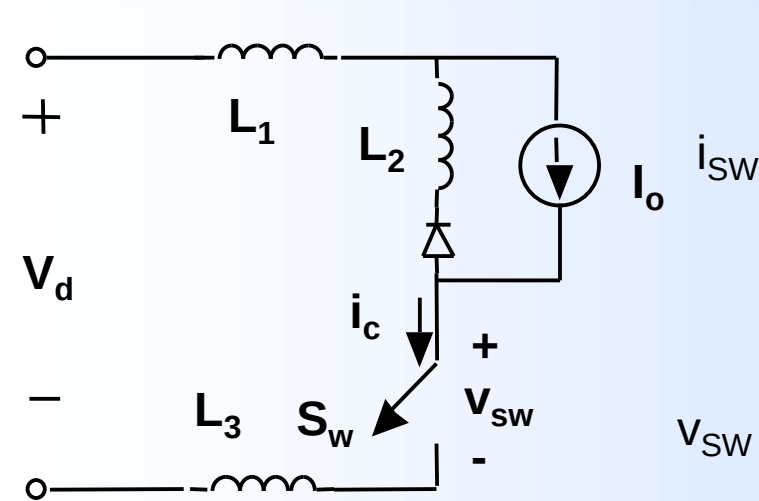


Fig. 27-11(a)

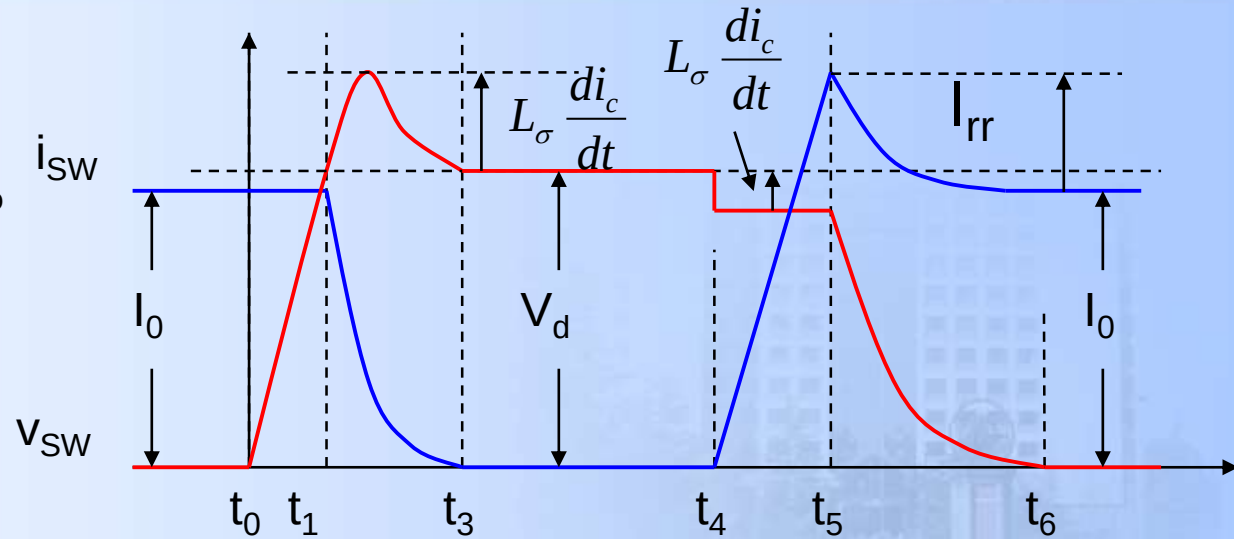


Fig. 27-11(c)

Turn-off: at $t=t_0$, the switch voltage begins to rise,

at $t=t_1$, the diode begins to conduct,

at $t=t_3$, the voltage comes down to V_d ,

Turn-on: at $t=t_4$, the switch current begins to rise,

at $t=t_5$, the freewheeling diode recovers,

at $t=t_6$, the switch voltage decreases to zero.

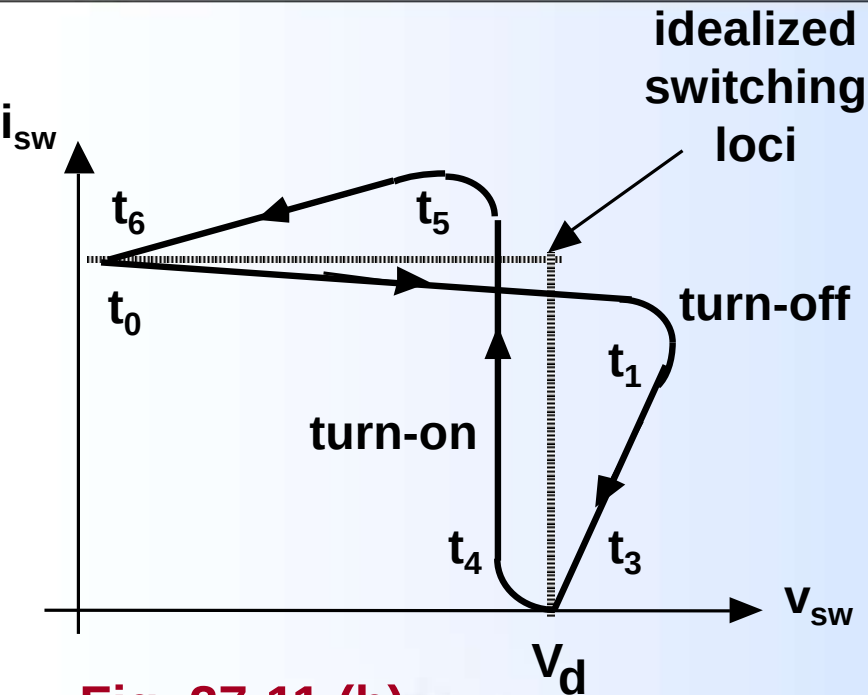


Fig. 27-11 (b)

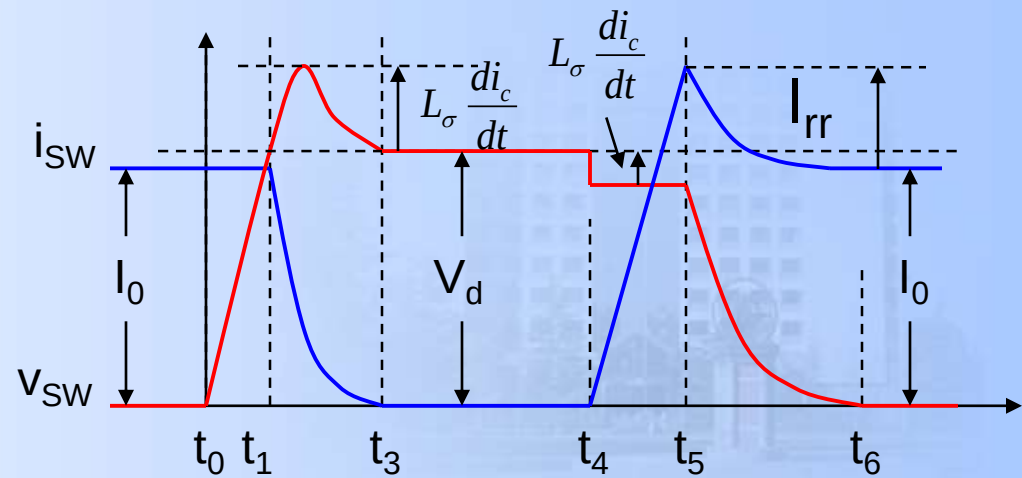


Fig. 27-11(c)

➤ Both voltage and current of the switch are high simultaneously at turn-on and turn-off, thus causing a high instantaneous power dissipation.

➤ The stray inductances result in overvoltage beyond V_d .

➤ The diode reverse-recovery current causes overcurrent beyond I_0 .

27-5 Turn-off Snubber for Controlled Switches ☆

Step-down converter with turn-off snubber

1 The **goal** of a turn-off snubber is to provide a zero (low) voltage across the switch while the current turns off.

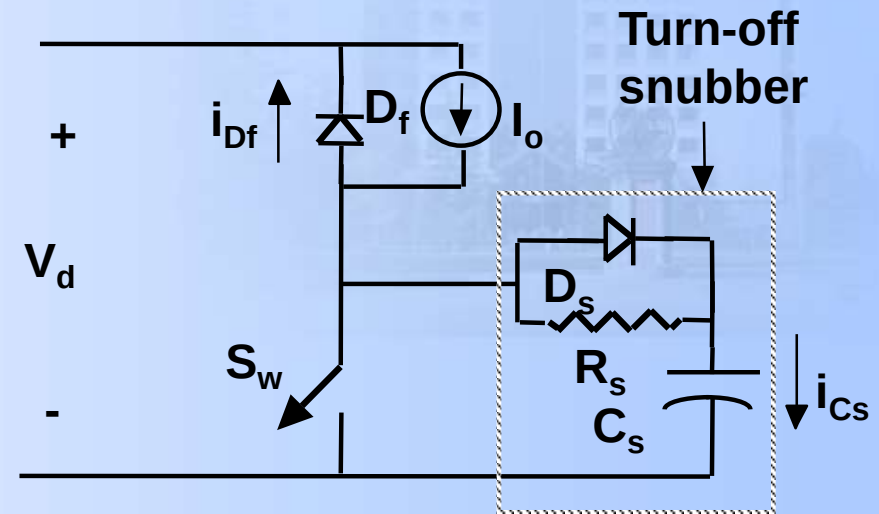


Fig. 27-12

(a)

2 RCD snubber 工作原理

- D_s shorts out R_s during S_w turn-off.
- During S_w turn-on, D_s reverse-biased and C_s discharges through R_s .

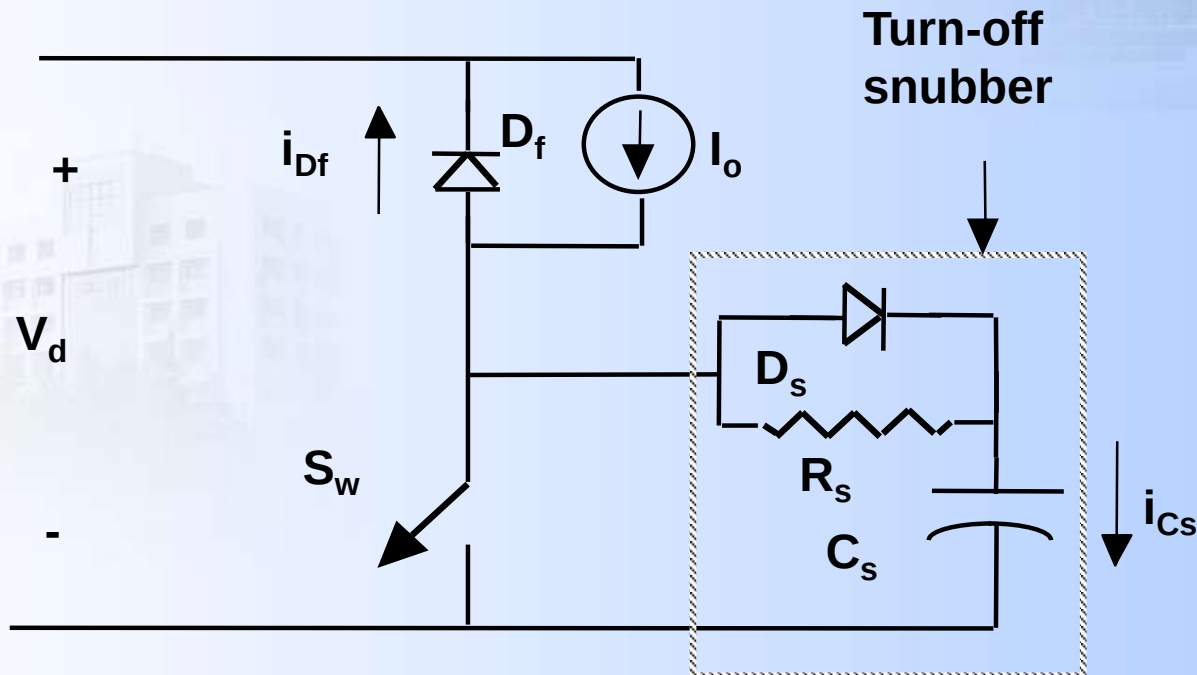


3 RCD snubber 各元件的作用

C_s 的作用是限制上 S_w 的 dv/dt 和限制关断过电压；

R_s 的作用是当 S_w 开通时泄放 C_s 上的能量，并限制放电电流；

D_s 的作用是当 S_w 关断时使 R_s 短路，更好发挥限制上 S_w 的 dv/dt 和限制关断过电压的作用。



4 RCD snubber 电路分析

(1) Equivalent circuit during switch turn-off.

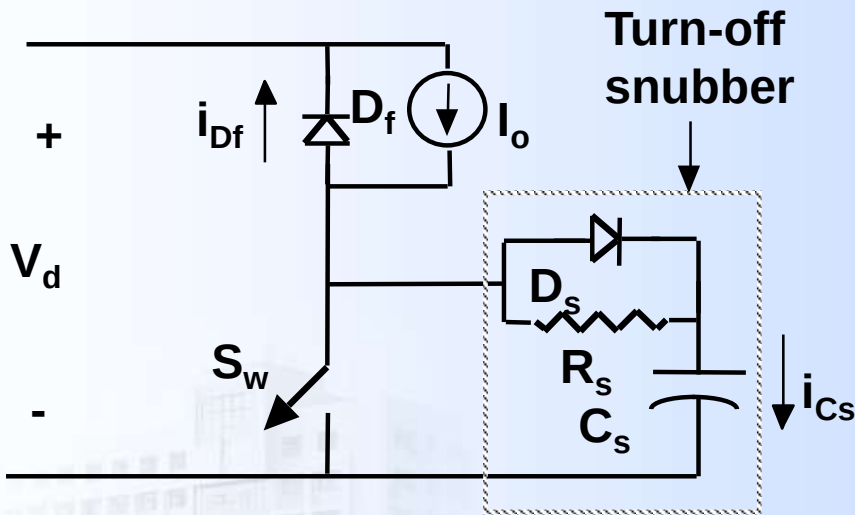


Fig. 27-12 (a)

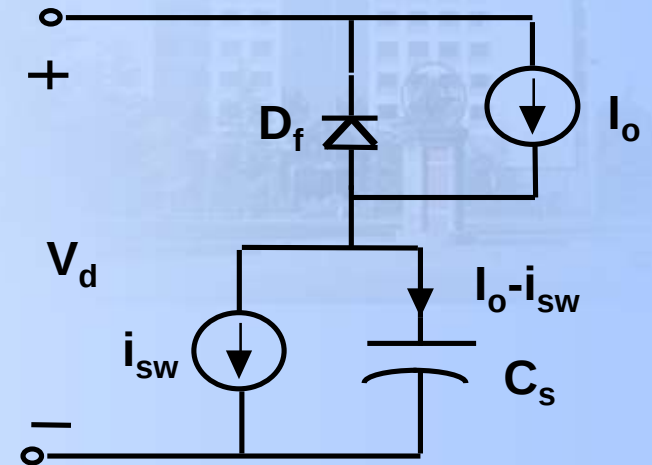


Fig. 27-12 (b)

(2) Circuit waveforms for various values of C_s

$$i_{sw} = I_o - \frac{I_o t}{t_{fi}}$$

$$v_{Cs} = v_{sw} = \frac{1}{C_s} \int_0^t i_{Cs} dt$$

$$= \frac{I_o t^2}{2C_s t_{fi}}$$

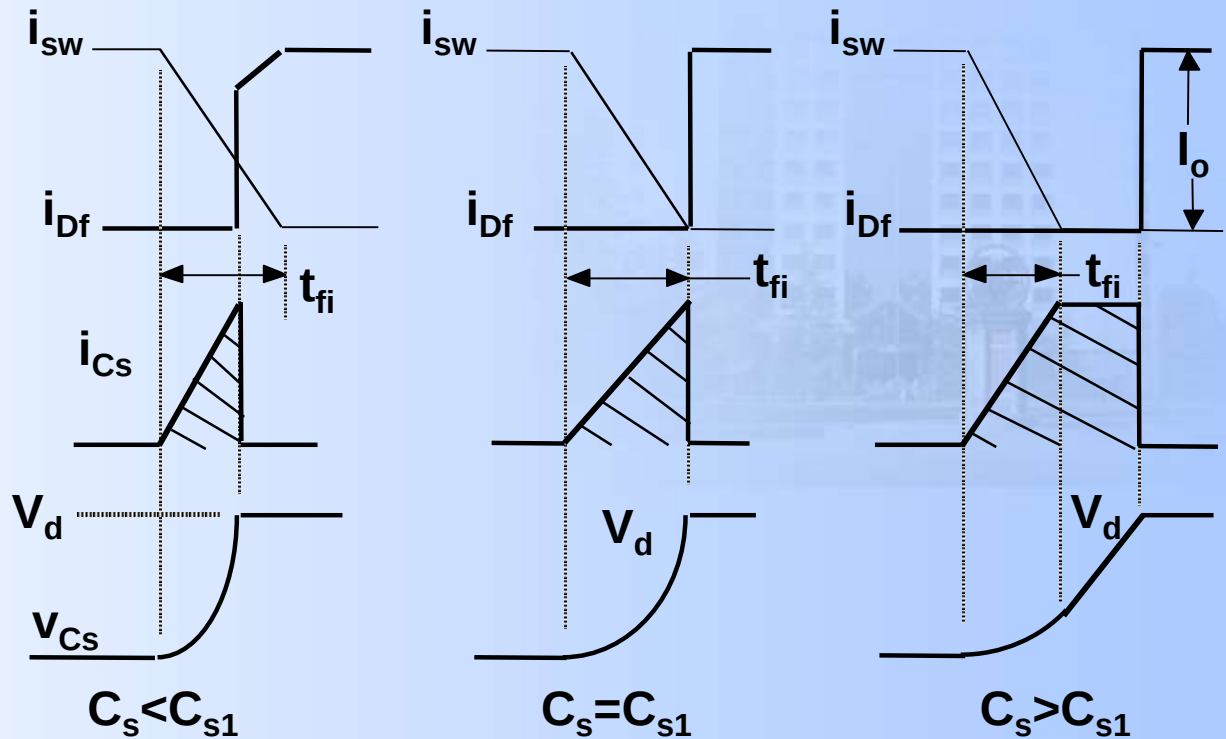


Fig. 27-12 (c)

➤ Simplifying assumptions

(1) No stray inductance

(2) $i_{sw}(t)$ unaffected by snubber circuit and changed linearly.

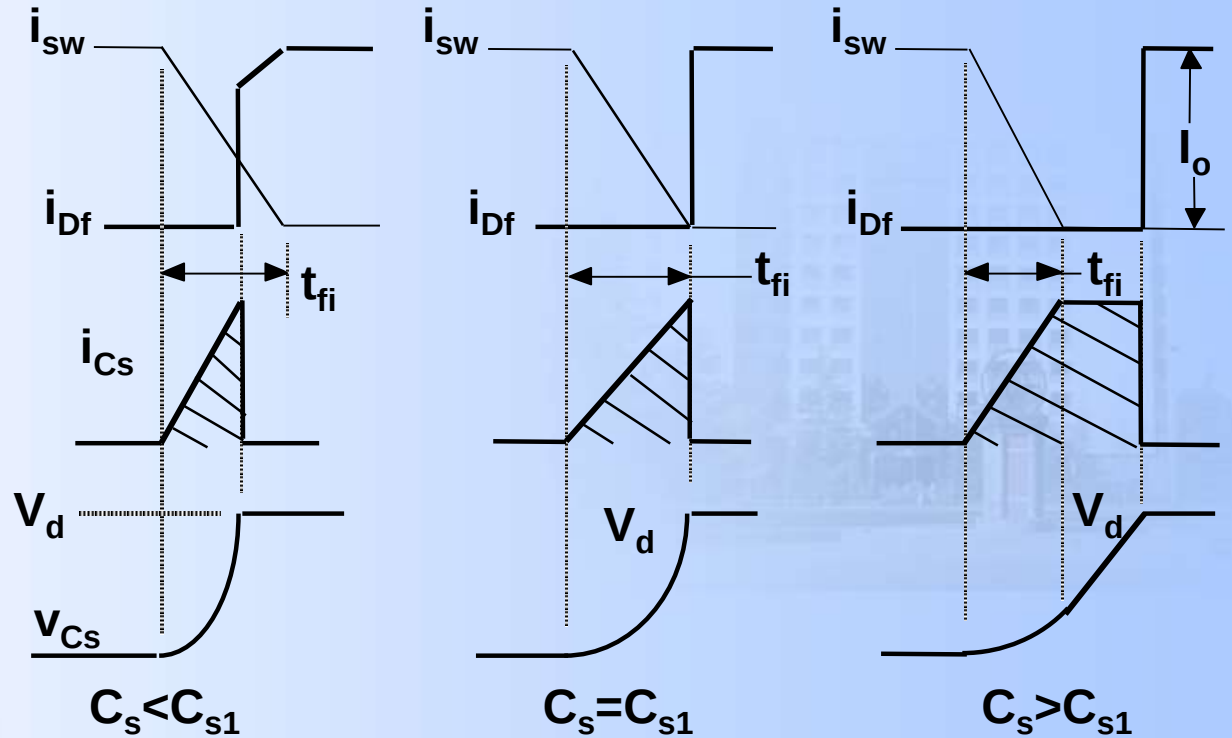


For a current
fall time t_{fi} ,
when $0 < t < t_{fi}$

$$C_{s1} = \frac{I_o t_{fi}}{2V_d}$$

$$w_{off} = \int_0^{t_{fi}} v_{sw} i_{sw} dt$$

$$P_{off} = w_{off} f_s$$



$$i_{Cs} = \frac{I_o t}{t_{fi}} \quad i_{sw} = I_o - \frac{I_o t}{t_{fi}} \quad (27-26)$$

$$v_{Cs} = v_{sw} = \frac{1}{C_s} \int_0^t i_{Cs} dt = \frac{I_o t^2}{2C_s t_{fi}} \quad (27-27) \quad 37$$



Turn-off Snubber for Controlled Switches

5 The turn-off Switching trajectory(关断轨迹)

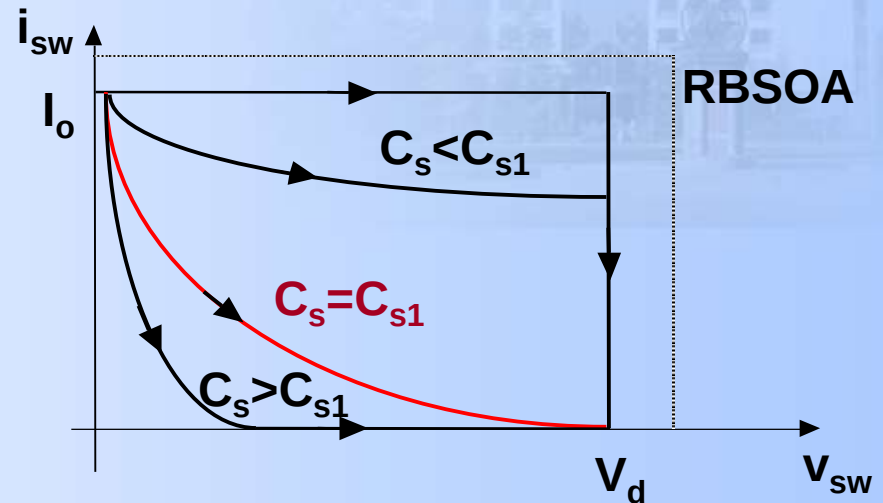
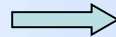
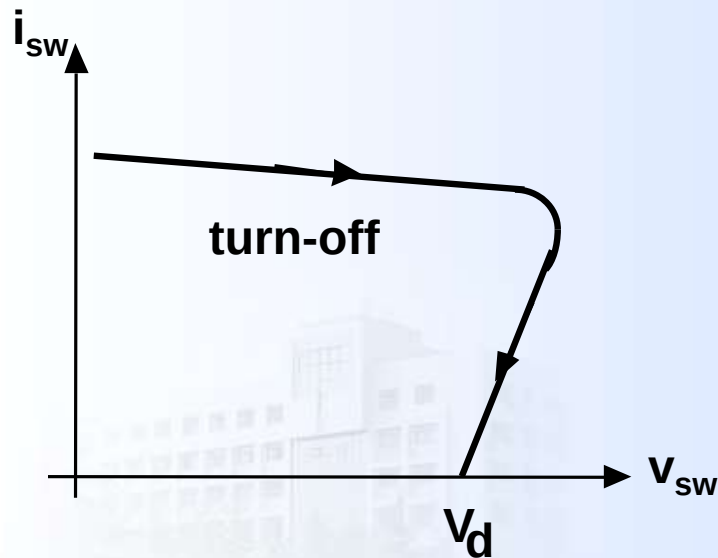


Fig. 27-13 Switching trajectory during turn-off with various values of snubber capacitance C_s .

27-6 Overvoltage Snubber

- Step-down converter with overvoltage snubber comprised of D_{ov} , C_{ov} , and R_{ov} .
- Overvoltage snubber limits overvoltage (due to stray Inductance) across S_w as it turns off.

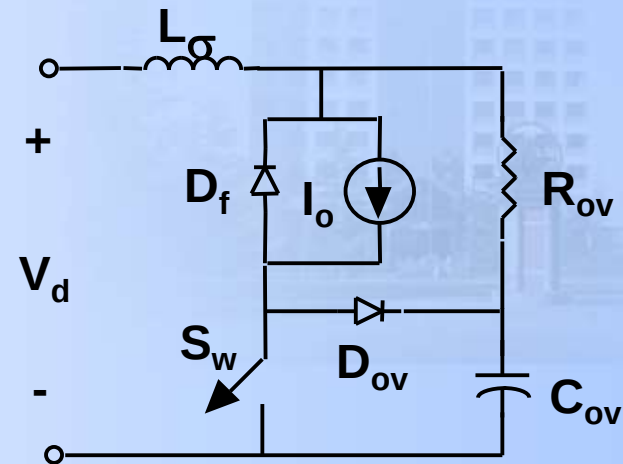


Fig. 27-16(a) Overvoltage snubber

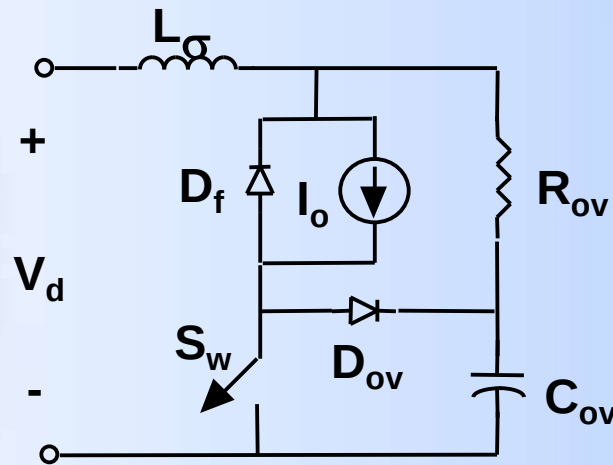


➤ The roles of elements

C_{ov} 的作用是限制 S_w 上的 dv/dt 和限制关断过电压；

R_{ov} 的作用泄放 C_{ov} 上的能量，并限制放电电流；

D_{ov} 的作用是提供 C_{ov} 充电回路，更好发挥限制 S_w 的 dv/dt 和限制关断过电压的作用。





27-7 Turn-on Snubber

1 Step-down converter with turn-on snubber

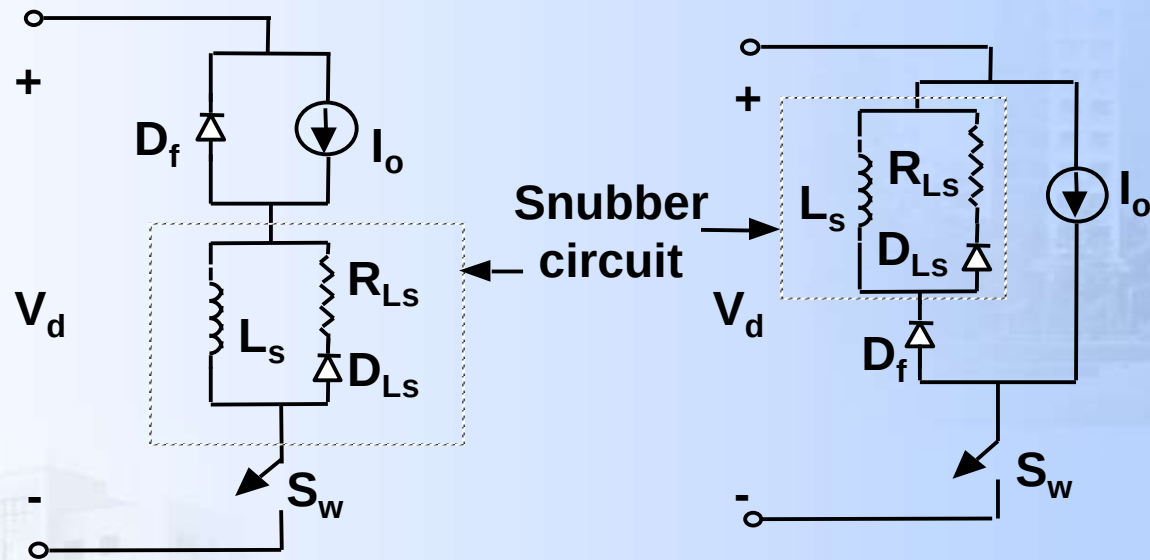
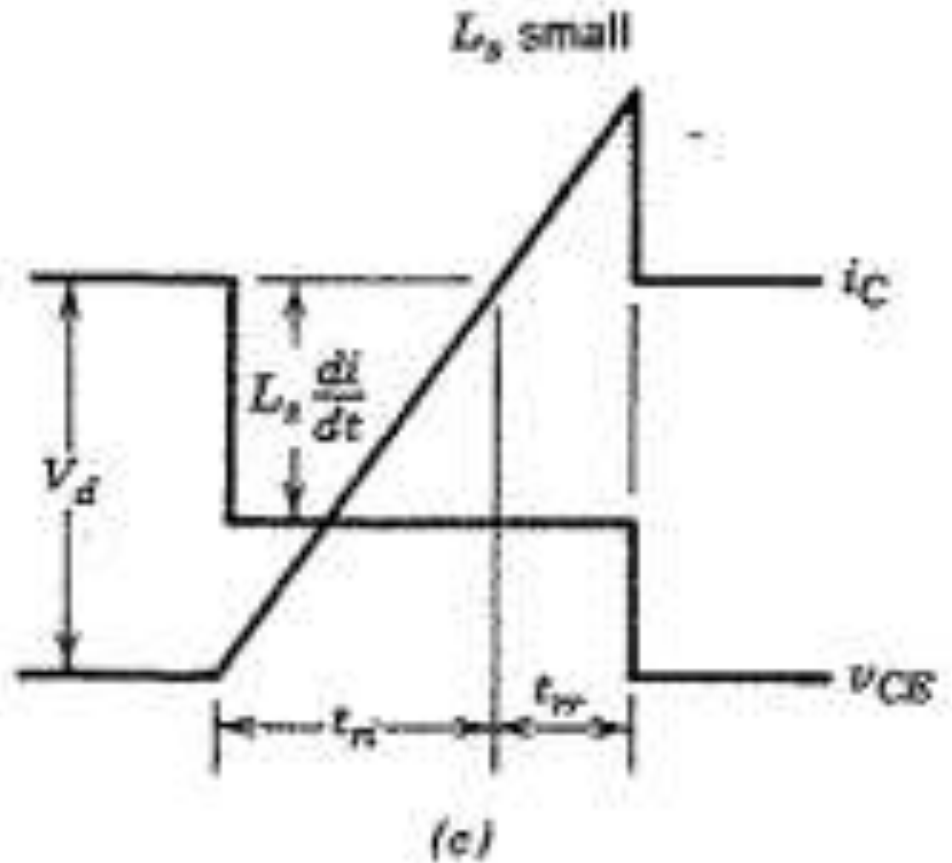
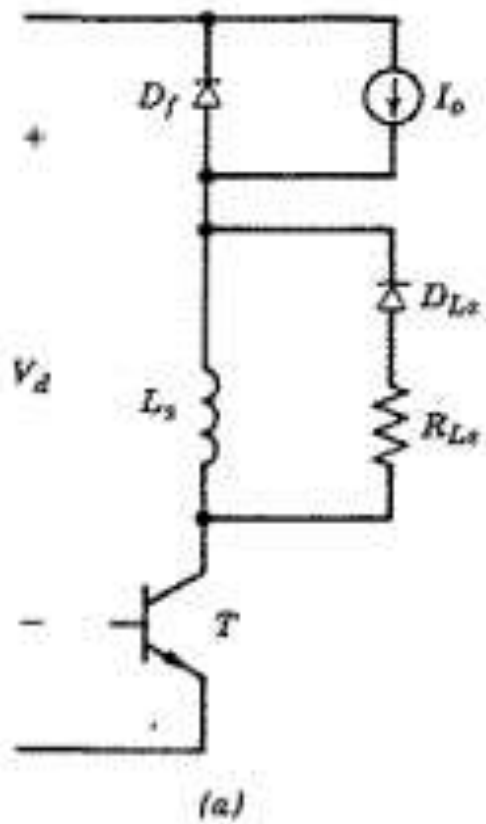
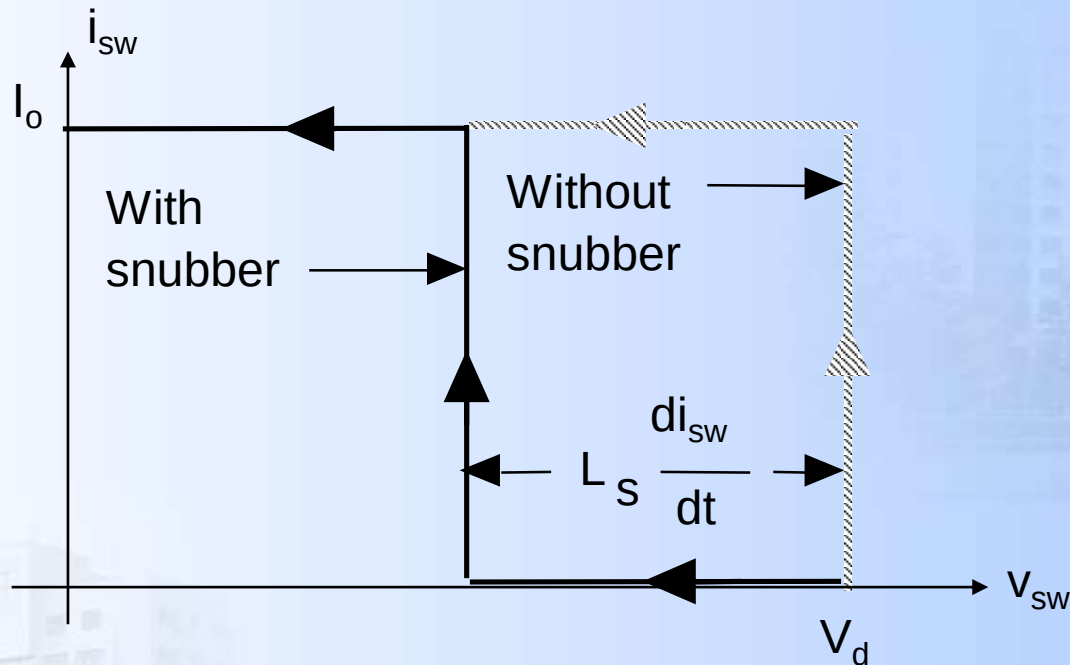


Fig. 27-17 (a) and (b)





2 Switching trajectory with and without turn-on snubber.



The reduction in the voltage across the switch during turn-on is:

$$\Delta V_{sw} = \frac{L_s I_o}{t_{ri}} \quad (27-39)$$



1. Snubber circuits protect semiconductor devices during turn-on and turn-off transients by limiting voltage and current magnitudes and rates of rise and by shaping the switching trajectory of the device as it turns on and off.
2. There are three topological snubber configurations: unpolarized RC snubbers, polarized RC snubbers, and polarized RL snubbers.
3. Unpolarized RC snubbers protect diodes and thyristors from overvoltages generated by the reverse-recovery current snap-off in the presence of stray series inductance.



4. A polarized RC circuit functions as turn-off Snubber to protect all types of controlled switches (IGBT, MOSFET, etc) by providing a low voltage across the switch while the current turns off. The turn-off switching trajectory traverses a low-power dissipation portion of the output I-V plane and the rate of rise (dv/dt) of the voltage across the device during turn-off is controlled.
5. A polarized RC circuit with a different topology than a turn-off snubber can be used as an overvoltage snubber to limit the overvoltage experienced by a controlled switch when it turns off in the presence of series inductance.



6. A polarized RL snubber protects controlled switches by reducing the voltage across a device as the current through it builds up during the turn-on transient. These turn-on snubbers can also be designed to limit overcurrents in the device during the turn-on transient due to diode reverse recovery.



1. What are the Snubber's Functions?
2. What is a polarized snubber?
3. What is a nonpolarized snubber?
4. What is the cause of reverse recovery transient voltages?
5. What is the purpose of di/dt protection?
6. What is the common method of di/dt protection?
7. What is the purpose of dv/dt protection?
8. What is the common method of dv/dt protection?
9. What is the turn-off characteristic of thyristors?